

OS LAB MANUAL

(CS23431)

Lab:3

Roll No:230701229

EX.NO:6b SHORTEST JOB

FIRST

Aim: To implement the Shortest Job First (SJF) scheduling technique

Program:

```
#include <stdio.h>

struct Process {

int id;

int burst_time;

int waiting_time;

int turnaround_time;

};

void sortProcesses(struct Process p[], int n) {

    struct Process temp;

    for (int i = 0; i < n - 1; i++) {

        for (int j = i + 1; j < n; j++) {

            if (p[i].burst_time > p[j].burst_time) {

                temp = p[i];

                p[i] = p[j];

                p[j] = temp;

            }

        }

    }

}

int main() {
```

```

int n;
struct Process p[10];
int total_waiting_time = 0, total_turnaround_time = 0;
printf("Enter the number of processes: ");
scanf("%d", &n);
printf("Enter the burst time of the processes:\n");
for (int i = 0; i < n; i++) {

    printf("Process %d: ", i + 1);
    scanf("%d", &p[i].burst_time);
    p[i].id = i + 1;
    p[i].waiting_time = 0;
    p[i].turnaround_time = 0;
}
sortProcesses(p, n);
for (int i = 0; i < n; i++) {
    if (i == 0) {
        p[i].waiting_time = 0;
    } else {
        p[i].waiting_time = p[i - 1].waiting_time + p[i - 1].burst_time;
    }
    p[i].turnaround_time = p[i].waiting_time + p[i].burst_time;
    total_waiting_time += p[i].waiting_time;
    total_turnaround_time += p[i].turnaround_time;
}
printf("\nProcess\tBurst Time\tWaiting Time\tTurn Around Time\n");
for (int i = 1; i < n; i++) {

    printf("%d\t%d\t%d\t%d\n", p[i].id, p[i].burst_time, p[i].waiting_time,
p[i].turnaround_time);
}
printf("\nAverage waiting time is: %.2f", (float)total_waiting_time / n);

```

```
*printf("\nAverage Turn Around Time is: %.2f\n", (float)total_turnaround_time / n);  
  
    return 0;  
  
}
```

Input:

```
Enter the number of processes: 4  
Enter the burst time of the processes:  
Process 1: 2  
Process 2: 3  
Process 3: 1  
Process 4: 4
```

OUTPUT:

```
Enter the number of processes: 4  
Enter the burst time of the processes:  
Process 1: 2  
Process 2: 3  
Process 3: 1  
Process 4: 4  
  
Process Burst Time      Waiting Time      Turn Around Time  
1         2             1                 3  
2         3             3                 6  
4         4             6                 10  
  
Average waiting time is: 2.50  
Average Turn Around Time is: 5.00
```